



Dr. Carlos Manuel Herrera Castillo

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RESEARCH EXPERIENCE

PhD research, Department of Environmental Sciences, Universität Basel. 06.2021 – 09.2025

- Objective: determining the genomic and transcriptomic causes of target phenotypes.
- Processed large-scale genomic datasets using high-performance computing clusters.
- Designed and executed complementation crosses to test candidate loci.
- Identified divergence stages via skeletal double-staining.
- Performed tissue sampling for bulk RNA-seq, single-cell RNA-seq, and single-cell ATAC-seq.
- Integrated multi-omics data to identify candidate regulatory elements and genes behind morphological trait loss.

MSc research, Paläontologisches Institut, Universität Zürich. 09.2019 – 06.2021

- **Project 1: Domestication and Development in Birds:**
- Investigated how domestication affects morphological development domestic birds.
- **Project 2: Description of New Fossil Material of *Laquintasaura venezuelae*:**
- Described newly recovered fossil material from an early ornithischian dinosaur (*Laquintasaura venezuelae*).

RELEVANT RESEARCH SKILLS

- Sample fixation, DNA and RNA extraction, PCR, electrophoresis QC, library preparation.
- Use of bioinformatics packages to process and interpret genomic, transcriptomic and multiomic (scRNAseq + ATACseq) data.
- Ion chromatography, enzymatic kinetics, Northern, Southern and Western Blotting.
- Whole-mount skeletal staining.
- 2D and 3D X-ray, 3D output processing, landmark selection, procrustes analyses for morphometrics.

SOFTWARES

R for genomic and transcriptomic processing, STAR align (RNAseq aligner), NovoAlign and bwa-mem2 (Genomic aligners), AlphaFold, Mimics (3D scan processing), HPC

EDUCATION

Universität Basel. PhD in Zoology (specialization in genomics and transcriptomics) 06.2021 – 09.2025

Universität Zürich. MSc in Biology 09.2019 – 06.2021

Universidad Autónoma de Madrid. Bachelors in Biology 09.2015 – 06.2019

PUBLICATIONS

Evolution of threespine stickleback dorsal spines via *hoxdb* gene regulation. 2025

Herrera-Castillo, Carlos Manuel, Brechbühl, Tanja, Fages, Antoine, Louise Dean, Laura, Tschopp, Patrick, MacColl, Andrew, Berner, Daniel 10.1101/2025.08.04.667941

Skeletal variation in bird domestication: limb proportions and sternum in chicken, with comparisons to mallard ducks and Muscovy ducks. 2022

Herrera-Castillo, Carlos Manuel, Geiger, Madeleine, Núñez-León, Daniel, Nagashima, Hiroshi, Gebhardt-Henrich, Sabine, Toscano, Michael, Sanchez-Villagra, Marcelo R. 10.7717/peerj.13229

A theropod trackway providing evidence of a pathological foot from the exceptional locality of Las Hoyas (upper Barremian, Serranía de Cuenca, Spain). 2022

Herrera-Castillo, Carlos Manuel, Moratalla, José J., Belaústegui, Zain, Marugán-Lobón, Jesús, Martín-Abad, Hugo, Nebreda, Sergio M., López-Archilla, Ana I., Nebreda, Sergio M., Nebreda, Sergio M., Buscalioni, Ángela D. 10.1371/journal.pone.0264406

Non-invasive imaging reveals new cranial element of the basal ornithischian dinosaur *Laquintasaura venezuelae*, Early Jurassic of Venezuela. 2021

Herrera-Castillo, Carlos Manuel, Carrillo-Briceño, Jorge D. Sánchez-Villagra, Marcelo R. 10.5167/uzh-208139

EXPECTED PUBLICATIONS

Developmental genetic basis of pelvic armor reduction in acidic-adapted threespine stickleback fish Expected: 08.2026

Herrera-Castillo, Carlos Manuel, Brechbühl, Tanja, Fages, Antoine, Tschopp, Patrick, MacColl, Andrew, Berner, Daniel

Assessing sticklebacks as potential vectors of *Pasteuria ramosa* parasitism in *Daphnia magna*: A case study on gut passage survival and transmission Expected: 08.2026

Lampadaridis, Nikolaos, *Herrera-Castillo, Carlos Manuel*, Ebert, Dieter

TEACHING EXPERIENCE

Lecturer: Introduction to Scientific Work, Universität Zürich, Switzerland 06.2025 – Present

Co-supervisor – Master's Thesis Project, Universität Basel, Switzerland 02.2024 – 02.2025

Lecturer: Introduction to Biology, Universität Basel, Switzerland 09.2024 – 12.2024

Introduction to laboratory protocols, Universität Basel, Switzerland 10.2023 – 04.2024

Lecture on Vertebrate Palaeozoology, Universität Zürich, Switzerland 09.2020 – 12.2020

CONFERENCES

Biology24 (University of Zürich - Switzerland) 2024

Oral presentation - Title: Investigating the causes of adaptation in three-spined sticklebacks.

3rd AsiaEvo Conference (National University of Singapore - Singapore) 2023

Oral presentation - Title: Investigating the causes of adaptation in three-spined sticklebacks (a spiny subject)

10th International Conference on Stickleback Behaviour and Evolution (University of Hólar - Iceland) 2022

Oral presentation - Title: Developmental genetic basis of skeletal evolution in stickleback fish.

Biology19 (University of Zürich - Switzerland) 2019

Poster presentation - Title: Skeletal variation in bird domestication.

AWARDS AND FUNDING

Freiwillige Akademische Gesellschaft fellowship 2025

The University of Basel travel fund for early career researchers 2023 and 2024

Scholarship from Swiss-European Mobility Programme (SEMP) 2019

CERTIFICATIONS AND LICENCES

LTK Module 20: Approved to perform animal experimentation on aquatic animals under Swiss law.

MEMBERSHIP OF PROFESSIONAL SOCIETIES

Member of the Society for the Study of Amphibians and Reptiles (SSAR)

Member of the Swiss Reproducibility Network

Reviewer in Journal *Molecular Ecology*

LANGUAGES

Languages: Spanish (native), French (family language), English (C2), German (B1)

REFEREES

Dr. Daniel Berner (PhD supervisor): Universität Basel, daniel.berner@unibas.ch

Prof. Dr. Marcelo Sánchez-Villagra (MSc supervisor): Universität Zürich, m.sanchez@pim.uzh.ch

Prof. Dr. Dieter Ebert (co-supervised MSc student with): Universität Basel, dieter.ebert@unibas.ch